

Nicholas Eisenberg, Ph.D.

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Experience

Senior Data Scientist

July 2022 - Present

Nevada National Security Site

- Contributed to a project employing SSD and YOLO for thermal image-based vehicle detection, reaching 80% mAP on in-house datasets. Deployed models using Streamlit, AWS, and Docker, providing easy client access via SSH and port forwarding.
- Played a key role in a project focused on anomaly detection in time-series data using a PyTorch-based LSTM autoencoder. Achieved 85% precision, aiding in predictive maintenance for pulse-powered X-ray machines.
- Sole developer of a Django-based web interface for Neo4j graph database interaction, simplifying database access for non-technical engineers. Implemented Python classes for CSV data conversion into database nodes and relationships.
- Developed 'TorchFace', a Python-based PyTorch interface, enhancing team efficiency in neural network operations. This tool standardized dataset creation, training, validation, and evaluation processes, and automatically organized logs and model state dictionaries.
- Innovated an in-house Python library leveraging subprocess module for seamless file transfer to/from AWS and S3 buckets, enhancing data management and cloud integration capabilities.

Graduate Teaching Assistant

January 2016 - June 2022

University of Nevada, Las Vegas and Auburn University

- Instructed sections in Mathematical Statistics, Real Analysis, Calculus I, II and III and Pre-Calculus
- Managed Auburn University's math tutoring center of 14 tutors

Technical Skills

Programming Languages: Python, SQL, Cypher, Bash, R, Lua, $\text{\LaTeX}$

Libraries: Pytorch, Scikit-Learn, Scipy, Pandas, Ssqlalchemy, Plotly, Streamlit, Django Subprocess

Developer Tools: Git, AWS, Unix/Linux, Vim, Tmux, Neo4j

Education

Auburn University - Ph.D.

June 2022

Research Area: Probability (Stochastic Processes and Partial Differential Equations)

Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S.

December 2015 / January 2018

Major: Applied Mathematics

Publications

- 1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). <https://doi.org/10.1007/s40072-022-00258-6>
- 2: Eisenberg, N., Chen, L. Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ with no drift term. Accepted November 2023 to Journal of Theoretical Probability. Preprint available at <https://arxiv.org/pdf/2209.04771v1.pdf>